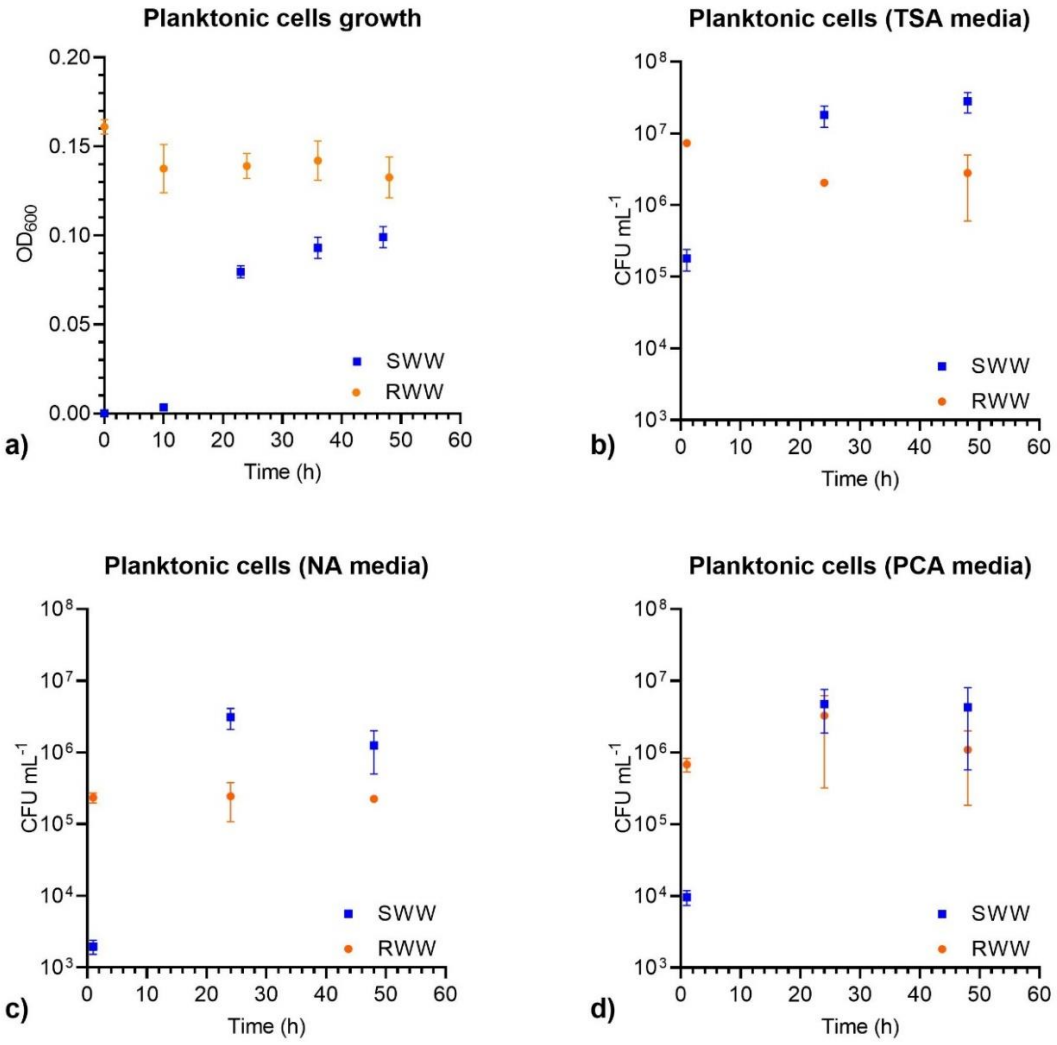
Resilience of Anodic Biofilm in Microbial Fuel Cell Biosensor for BOD Monitoring of Urban Wastewater

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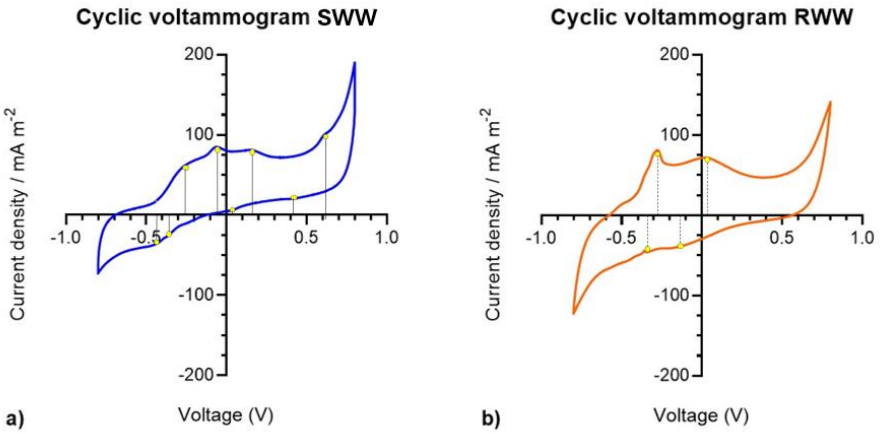
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Supplementary Figures

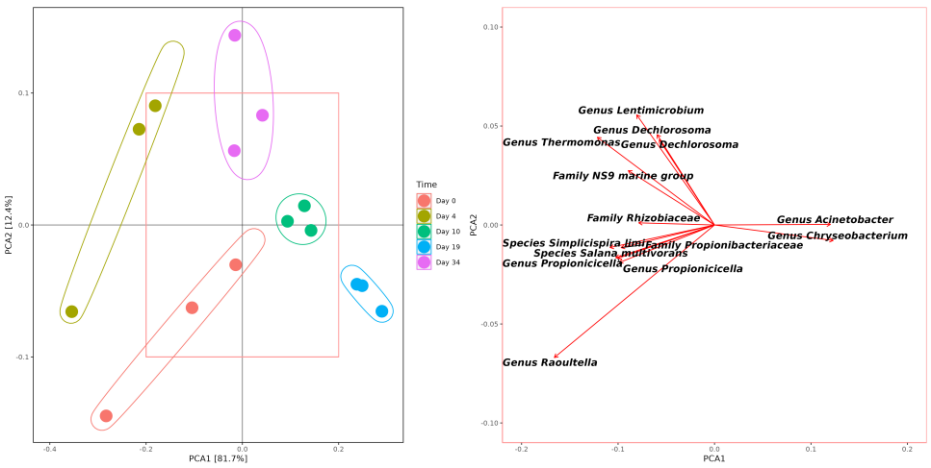


Supplementary Figure 1: The growth of planktonic cells in the anolyte was monitored in two MFC reactors, one fed with SWW (COD = 442 ± 8 mg mL^{-1}, pH = 7.2 ± 0.1, Conductivity =

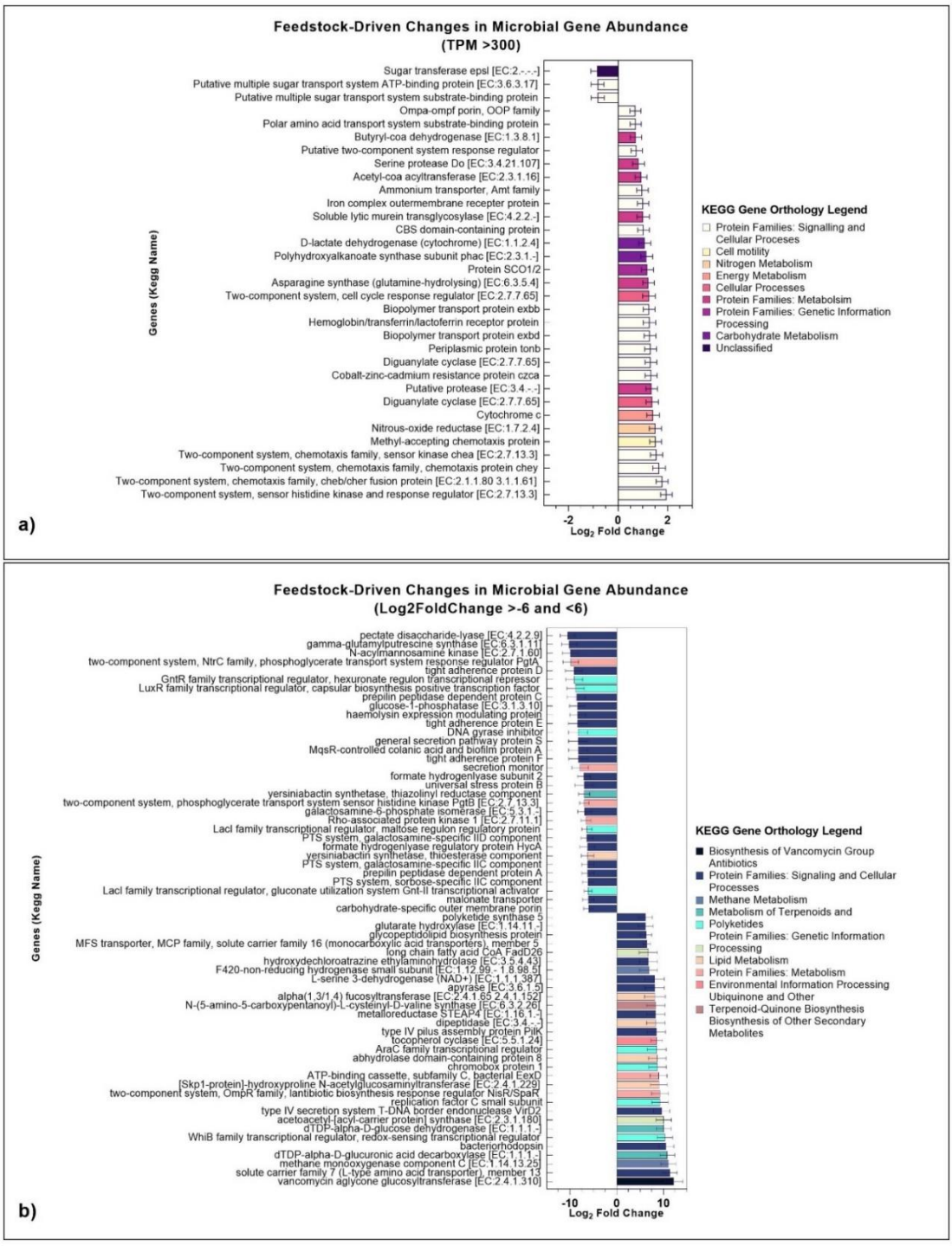
1.2 ± 0.1 mS cm^{-1}) and then with real wastewater RWW (COD = 602 ± 7 mg mL^{-1}, pH = 7.3 ± 0.1, Conductivity = 1.4 ± 0.1 mS cm^{-1}). Cell proliferation was estimated using a) the OD_{600} method, and plate count of total aerobic heterotrophic culturable bacteria in b) Tryptic Soy Agar (TSA), c) Nutrient Agar (NA), and d) Plate Count Agar (PCA). Samples of the anolyte were taken at time 0 (after few minutes from anolyte replacement) and at various time intervals over a period of two days, which corresponds to the maximum duration required to complete a peak current production.



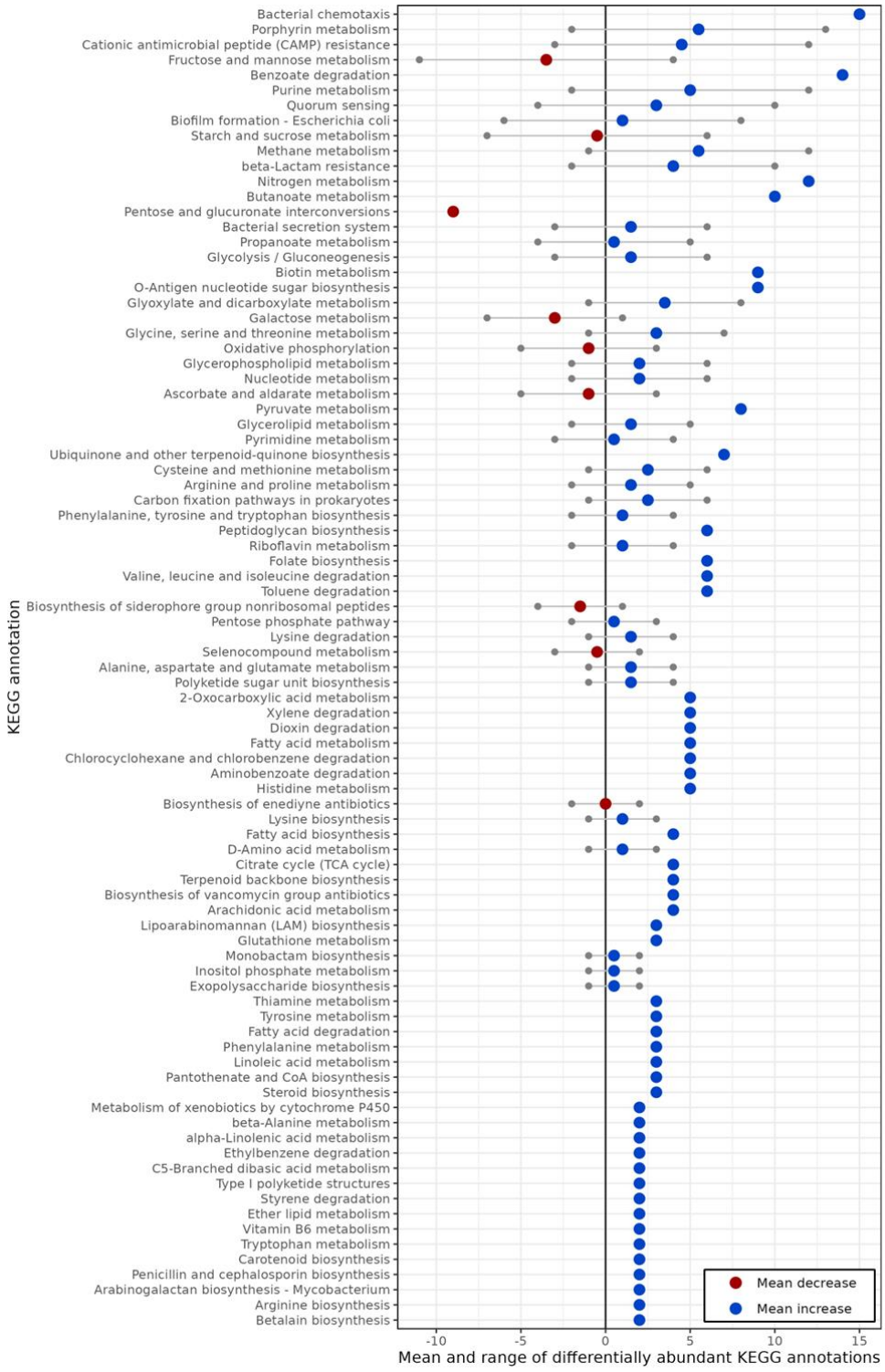
Supplementary Figure 2: Cyclic voltammograms of (a) MFC fed for over one month with SWW and (b) MFC fed for over one month with RWW with the oxidation and reduction peaks highlighted with yellow dots. The experiment was performed in the presence of oxidizable substrate (TO conditions, dotted line), which was either SWW (COD = 442 mg mL^{-1}, pH = 7.29, Conductivity = 1.31 mS cm^{-1}) or RWW (COD = 169 mg mL^{-1}, pH = 7.39, Conductivity = 1.31 mS cm^{-1}) and in absence of oxidizable substrate (NTO conditions, continuous line), with Phosphate Buffer Saline solution (8 g L^{-1} NaCl; 0.2 g L^{-1} KCl; 1.44 g L^{-1} Na_2HPO_4; 0.14 g L^{-1} KH_2PO_4; pH=7.4).The analysis was performed using a PalmSense4 potentiostat connected in a three-configuration mode (anode as WE, cathode CE, and Ag/AgCl RE) at 0.5 mV s^{-1}.



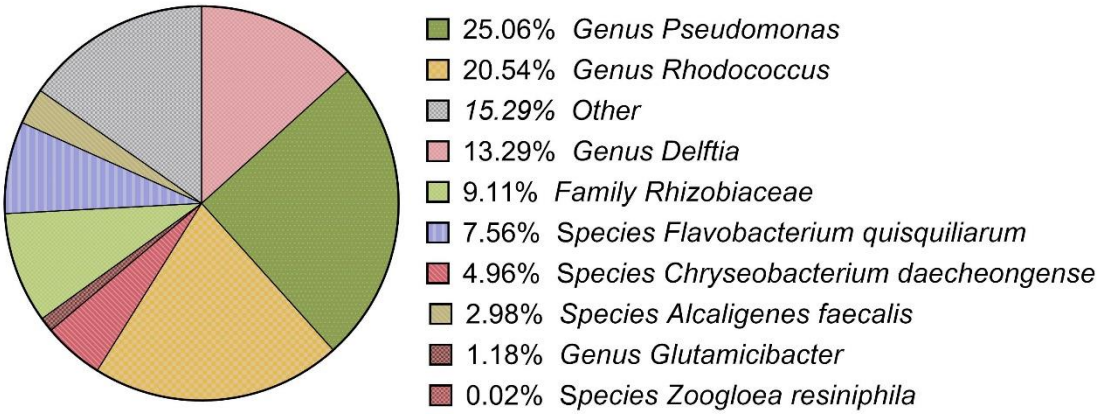
Supplementary Figure 3: Principal Coordinate Analysis of three replicates at day 0, 4,10, 19 and 34. The analysis, performed with the R package "phyloseq," is supplemented with weighted UniFrac distance assessment. The taxa were filtered above 0.025%.



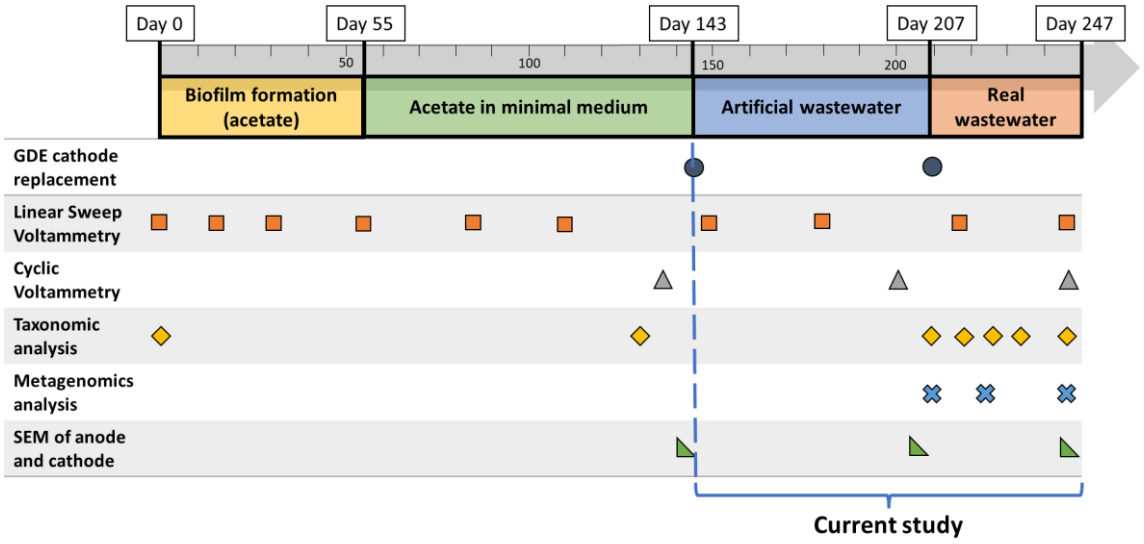
Supplementary Figure 4: Results of differential expression analysis applied to the KEGG-annotated metagenome, expressed in Log_2(Fold Change) at two levels of filtering a) Significant differentially abundant KEGG annotations (adjusted p-value < 0.01) exceeding 300 genes per million (GPM), and b) significant differentially abundant KEGG annotations (adjusted p-value < 0.01) exhibiting the greatest Log_2(Fold Change) upon transitioning from SWW to RWW. Negative values indicate a decrease in gene abundance when moving from SWW to RWW, whereas positive values signify an increase.



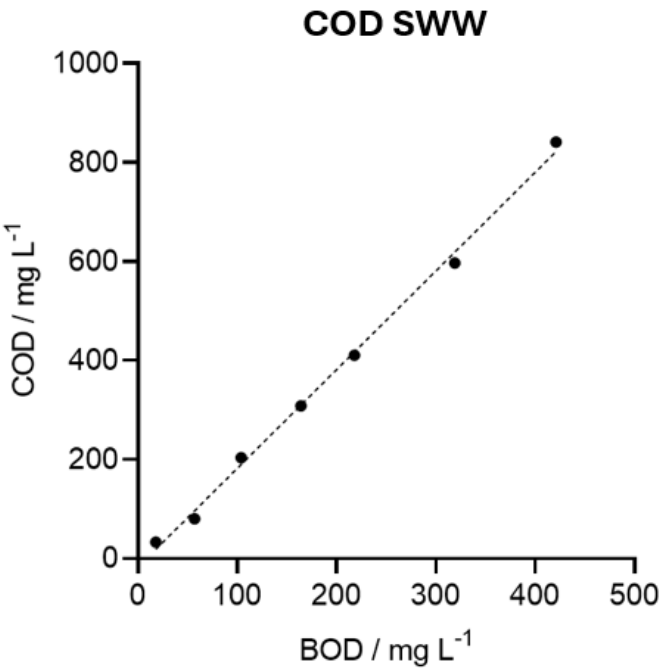
Supplementary Figure 5: Differential abundance of KEGG-annotated genes in anodic biofilms subjected to RWW and SWW feeding. Values plotted on the positive side of the x-axis represent an increased presence of genes under the RWW feeding. In contrast, annotations on the negative x-axis highlight a decreased gene abundance, indicating these genes were more prevalent in biofilms fed with SWW. The mean values are represented by coloured dots: a blue dot indicates an overall increase in the specific annotation after biofilms were fed with RWW. In contrast, a red dot signifies an overall decrease in the KEGG-annotated genes.



Supplementary Figure 6: Relative abundance of the lowest taxonomic rank assigned (species, genus and families) of the inoculum used in the study. The inoculum used in this study was sourced from an acetate-fed MFC, which had been in operation for more than a year and was initially inoculated with sediment from the campus pond (University of Surrey, Guildford, UK). Taxa with abundances exceeding 1% are individually displayed, while those with lower abundances are collectively categorized as "other."



Supplementary Figure 7: Visual representation of the timeline of the MFC reactors, showcasing the diverse interventions and analyses undertaken. These include replacement of the GDE cathode, linear sweep voltammetry, cyclic voltammetry, taxonomic and metagenomics analyses of the biofilm, and scanning electron microscopy of the anode and cathode. The current study focuses on the last two phases of the experiment using artificial wastewater and real wastewater as feeding solutions.



Supplementary Figure 8: Correlation between the BOD of the artificial wastewater (SWW) and its COD. The BOD/COD ratio is 0.51 ± 0.05 and is equal to the reciprocal of the slope. Data are reported as mean of 2 replicates ± SD (error bars are smaller than the dots).

Supplementary Tables

Supplementary Table 1: Maximum current density obtained in three MFC replicates (R1, R2, R3) fed with RWW and SWW with different BOD and COD concentrations. The variation in cell colours (grey or white) indicates groups with comparable BOD (or COD) values, which are depicted as mean values with corresponding error bars on the x-axis and y-axis in Figure 1.

RWW					SWW				
COD / mg L^{-1}	Calculated BOD / mg L^{-1}	Current density / mA m^{-2}			COD / mg L^{-1}	Calculated BOD / mg L^{-1}	Current density / mA m^{-2}		
		R1	R2	R3			R1	R2	R3
658	165	103	60	80	411	210	154	188	179
655	164	81	33	75	416	212	150	-	177
612	153	80	53	62	425	217	131	156	121
348	84	15	11	13	597	304	153	180	186
361	87	13	8	11	621	317	148	137	177
297	71	14	10	12	605	309	151	178	178
238	57	2	3	2	207	106	73	86	90
284	68	1	1	1	202	103	72	93	90
277	66	1	2	1	198	101	76	91	88
381	91	25	24	29	858	438	190	202	194
359	86	24	24	32	853	435	235	246	261
412	99	25	32	24	843	430	180	214	193
712	171	115	92	100	114	58	44	64	70
685	164	103	83	-	108	55	57	66	69
732	176	97	101	102	113	58	45	64	71
482	116	48	51	43	314	160	109	129	136
458	110	35	51	56	319	163	107	140	135
473	114	41	54	47	311	159	114	136	132
					1060	541	163	188	206
					1071	546	177	208	222
					1056	539	164	161	215

- :Data not available due to equipment fault.

Supplementary Table 2: Response time obtained in three MFC replicates (R1, R2, R3) fed with RWW and SWW with different BOD and COD concentrations.

RWW					SWW				
COD / mg L^{-1}	Calculated BOD / mg L^{-1}	Response time / h			COD / mg L^{-1}	Calculated BOD / mg L^{-1}	Response time / h		
		R1	R2	R3			R1	R2	R3
658	165	6.1	3.0	2.9	411	210	8.1	1.1	8.6
655	164	4.8	6.0	4.6	416	212	7.7	-	7.4
612	153	6.6	3.5	3.5	425	217	13.1	9.4	13.1

348	84	5.8	5.0	5.8	597	304	12.2	12.4	18.3
361	87	4.1	4.6	4.6	621	317	12.1	12.1	18.2
297	71	4.8	4.3	5.0	605	309	12.0	11.1	12.9
238	57	3.4	4.8	3.6	207	106	9.3	8.5	11.3
284	68	4.6	4.8	4.8	202	103	10.2	10.2	7.9
277	66	6.2	4.8	4.8	198	101	9.2	10.5	10.3
381	91	4.1	4.1	7.2	858	438	14.9	14.1	13.7
359	86	2.6	3.1	1.9	853	435	21.6	18.9	20.8
412	99	6.2	6.2	3.1	843	430	19.9	10.9	20.0
712	171	6.0	6.0	2.9	114	58	2.2	3.5	2.7
685	164	6.2	3.1	-	108	55	4.5	4.6	4.6
732	176	6.5	6.5	3.4	113	58	2.2	3.5	2.7
482	116	2.6	3.1	1.9	314	160	9.3	8.6	11.3
458	110	2.9	3.8	3.1	319	163	10.2	10.3	7.9
473	114	2.4	3.1	2.4	311	159	9.3	10.5	10.3
					1060	541	17.0	21.8	26.8
					1071	546	17.0	21.8	26.8
					1056	539	14.1	15.3	17.6

- :Data not available due to equipment fault.

Supplementary Table 3: List of the most abundant genes in both artificial wastewater (SWW) and real wastewater (RWW) samples, focusing on those exceeding 300 Transcripts Per Million (TPM) in either type of sample. Each gene is identified by a Kegg identifier (KO), Name, Kegg Orthology, Transcripts Per Million of the gene in SWW and RWW (expressed as mean of 2 biological replicates ± Standard Error of the Mean), and Log_2(Fold Change).

KO	Kegg Name	Kegg Orthology	TPM (SWW)	TPM (RWW)	Log2(Fold Change)
K11527	Two-component system, sensor histidine kinase and response regulator [EC:2.7.13.3]	Protein families: signaling and cellular processes	132 ± 18	545 ± 161	1.95
K13924	Two-component system, chemotaxis family, cheb/cher fusion protein [EC:2.1.1.80 3.1.1.61]	Protein families: signaling and cellular processes	439 ± 91	1618 ± 524	1.78
K03413	Two-component system, chemotaxis family, chemotaxis protein chey	Protein families: signaling and cellular processes	204 ± 46	677 ± 235	1.66
K03407	Two-component system, chemotaxis family, sensor kinase chea [EC:2.7.13.3]	Protein families: signaling and cellular processes	104 ± 20	368 ± 112	1.55
K03406	Methyl-accepting chemotaxis protein	Cell motility	650 ± 97	2168 ± 679	1.52
K00376	Nitrous-oxide reductase [EC:1.7.2.4]	Nitrogen metabolism	94 ± 24	302 ± 97	1.51
K08738	Cytochrome c	Energy metabolism	157 ± 37	440 ± 141	1.41
K13590	Diguanylate cyclase [EC:2.7.7.65]	Cellular Processes	154 ± 35	459 ± 155	1.38

K08303	Putative protease [EC:3.4.-.-]	Protein families: metabolism	103 ± 18	305 ± 70	1.36
K15726	Cobalt-zinc-cadmium resistance protein czca	Protein families: signaling and cellular processes	143 ± 36	430 ± 138	1.33
K21023	Diguanylate cyclase [EC:2.7.7.65]	Cellular Processes	441 ± 56	1239 ± 367	1.33
K03832	Periplasmic protein tonb	Protein families: signaling and cellular processes	170 ± 36	450 ± 121	1.32
K03559	Biopolymer transport protein exbd	Protein families: signaling and cellular processes	127 ± 27	330 ± 74	1.29
K16087	Hemoglobin/transferrin/lactoferrin receptor protein	Protein families: signaling and cellular processes	113 ± 37	331 ± 113	1.27
K03561	Biopolymer transport protein exbb	Protein families: signaling and cellular processes	115 ± 26	303 ± 74	1.25
K02488	Two-component system, cell cycle response regulator [EC:2.7.7.65]	Cellular Processes	139 ± 34	365 ± 124	1.25
K01953	Asparagine synthase (glutamine-hydrolysing) [EC:6.3.5.4]	Protein families: metabolism	141 ± 26	388 ± 93	1.23
K07152	Protein SCO1/2	Protein families: genetic information processing	142 ± 29	351 ± 101	1.19
K03821	Polyhydroxyalkanoate synthase subunit phac [EC:2.3.1.-]	Carbohydrate metabolism	126 ± 29	328 ± 88	1.16
K00102	D-lactate dehydrogenase (cytochrome) [EC:1.1.2.4]	Carbohydrate metabolism	173 ± 36	417 ± 126	1.08
K07182	CBS domain-containing protein	Unclassified: signaling and cellular processes	240 ± 17	543 ± 125	1.03
K08309	Soluble lytic murein transglycosylase [EC:4.2.2.-]	Protein families: metabolism	134 ± 21	337 ± 76	1.02
K02014	Iron complex outermembrane recepter protein	Protein families: signaling and cellular processes	136 ± 25	311 ± 106	1.02
K03320	Ammonium transporter, Amt family	Protein families: signaling and cellular processes	484 ± 56	1115 ± 331	1.01
K00632	Acetyl-coa acyltransferase [EC:2.3.1.16]	Lipid metabolism	322 ± 73	724 ± 214	0.98
K04771	Serine protease Do [EC:3.4.21.107]	Protein families: metabolism	153 ± 35	320 ± 87	0.93
K07814	Putative two-component system response regulator		180 ± 43	358 ± 95	0.83
K00248	Butyryl-coa dehydrogenase [EC:1.3.8.1]	Lipid metabolism	483 ± 2	882 ± 169	0.75
K02030	Polar amino acid transport system substrate-binding protein	Protein families: signaling and	174 ± 46	306 ± 8	0.72

		cellular processes			
K03286	Ompa-ompf porin, OOP family	Protein families: signaling and cellular processes	495 ± 105	928 ± 206	0.71
K10546	Putative multiple sugar transport system substrate-binding protein	Protein families: signaling and cellular processes	177 ± 25	328 ± 62	0.70
K10548	Putative multiple sugar transport system ATP-binding protein [EC:3.6.3.17]	Protein families: signaling and cellular processes	383 ± 45	222 ± 28	-0.83
K19428	Sugar transferase epsl [EC:2.-.-.-]	Protein families: signaling and cellular processes	417 ± 55	232 ± 31	-0.84
K11527	Two-component system, sensor histidine kinase and response regulator [EC:2.7.13.3]	Protein families: signaling and cellular processes	333 ± 45	206 ± 40	-0.84

Supplementary Table 4: List of genes with the most pronounced log_2(fold change), greater than 6 or less than -6. on those exceeding 300 Transcripts Per Million in either type of sample. Each gene is identified by a Kegg identifier (KO), Name, Kegg Orthology, TPM of the gene in SWW and RWW (expressed as mean of 2 biological replicate ± Standard Error of the Mean), and Log_2(Fold Change).

Ko	Kegg Name	Kegg Orthology	TPM (SWW)	TPM (RWW)	Log2(Fold Change)
K16444	Vancomycin aglycone glucosyltransferase [ec:2.4.1.310]	Biosynthesis of vancomycin group antibiotics	0 ± 0	9.9 ± 5.2	12.20
K09285	Ap2-like factor, ant lineage	Protein families: signaling and cellular processes	0 ± 0	3.9 ± 1.8	11.43
K13870	Solute carrier family 7 (l-type amino acid transporter), member 13	Methane metabolism	0 ± 0	6.5 ± 3.4	11.33
K16161	Methane monooxygenase component c [ec:1.14.13.25]	Metabolism of terpenoids and polyketides	0 ± 0	3.9 ± 1.8	11.06
K18886	Gibberellin a4 carboxyl methyltransferase [ec:2.1.1.276]	Protein families: signaling and cellular processes	0 ± 0	5.2 ± 2.2	10.97
K21817	Beta-carotene 15,15'-dioxygenase [ec:1.13.11.63]	Protein families: genetic information processing	0 ± 0	7.1 ± 2.6	10.96
K21332	Dtdp-alpha-d-glucuronic acid decarboxylase [ec:1.1.1.-]	Metabolism of terpenoids and polyketides	0 ± 0	5 ± 2.4	10.79
K18961	T4 rna ligase 1 [ec:6.5.1.3]	Lipid metabolism	0 ± 0	2.8 ± 0.1	10.76
K18942	T4virus DNA polymerase [ec:2.7.7.7]	Protein families: signaling and cellular processes	0 ± 0	2.8 ± 1.2	10.75

K04641	Bacteriorhodopsin	Protein families: genetic information processing	0 ± 0	1.7 ± 0.3	10.59
K18956	Whib family transcriptional regulator, redox-sensing transcriptional regulator	Environmental Information Processing	0 ± 0	1.1 ± 0.2	10.32
K21079	Major capsid protein	Protein families: metabolism	0 ± 0	1.3 ± 0.3	10.07
K21331	Dtdp-alpha-d-glucose dehydrogenase [ec:1.1.1.-]	Environmental Information Processing	0 ± 0	1.3 ± 0.1	10.07
K18473	Acetoacetyl-[acyl-carrier protein] synthase [ec:2.3.1.180]	Protein families: genetic information processing	0 ± 0	2.5 ± 0.9	10.04
K21523	Terminase, large subunit [ec:3.1.-.-]	Protein families: metabolism	0 ± 0	1.1 ± 0.2	9.77
K18950	Ribonuclease h [ec:3.1.26.4]	Protein families: genetic information processing	0 ± 0	0.8 ± 0.6	9.71
K18960	Recombination endonuclease vii	Ubiquinone and other terpenoid-quinone biosynthesis	0 ± 0	5.7 ± 2.5	9.69
K18434	Type iv secretion system t-DNA border endonuclease virD2	Protein families: signaling and cellular processes	0 ± 0	0.9 ± 0.1	9.59
K16192	Uncharacterized membrane protein arfb	Protein families: metabolism	0 ± 0	1.3 ± 0.1	9.33
K04801	Replication factor c small subunit	Protein families: genetic information processing	0 ± 0	1.6 ± 0.3	9.25
K20488	Two-component system, ompr family, lantibiotic biosynthesis response regulator nisr/spar	Biosynthesis of other secondary metabolites	0 ± 0	0.8 ± 0.3	9.25
K12244	[skp1-protein]-hydroxyproline n-acetylglucosaminyltransferase [ec:2.4.1.229]	Protein families: metabolism	0 ± 0	0.8 ± 0.8	9.03
K16299	Atp-binding cassette, subfamily c, bacterial eexd	Nucleotide metabolism	0 ± 0	1.8 ± 0.7	8.95
K11585	Chromobox protein 1	Unclassified: metabolism	0 ± 0	1.3 ± 0.8	8.74
K13701	Abhydrolase domain-containing protein 8	Methane metabolism	0.1 ± 0.1	4 ± 2	8.67
K11920	Arac family transcriptional regulator	Xenobiotics biodegradation and metabolism	0 ± 0	0.9 ± 0.1	8.58
K09834	Tocopherol cyclase [ec:5.5.1.24]	Lipid biosynthesis proteins	0 ± 0	0.9 ± 0.4	8.55
K02661	Type iv pilus assembly protein pilk	Protein families: signaling and cellular processes	0 ± 0	3.9 ± 2	8.48
K08659	Dipeptidase [ec:3.4.-.-]	Unclassified: metabolism	0.6 ± 0.5	33.7 ± 14.7	8.36
K19876	Metalloreductase steap4 [ec:1.16.1.-]	Amino acid metabolism	0 ± 0	1.4 ± 0.6	8.35

K12743	N-(5-amino-5-carboxypentanoyl)-l-cysteinyl-d-valine synthase [ec:6.3.2.26]	Lipid biosynthesis proteins	0.5 ± 0.5	41.4 ± 18.7	8.31
K10953	Rtx toxin rtxa	Protein families: signaling and cellular processes	1.3 ± 0.4	0 ± 0	8.31
K20151	Alpha(1,3/1,4) fucosyltransferase [ec:2.4.1.65 2.4.1.152]	Protein families: signaling and cellular processes	1.6 ± 0.5	0 ± 0	8.17
K14641	Apyrase [ec:3.6.1.5]	Carbohydrate metabolism	3 ± 0.8	0 ± 0	8.14
K18962	T4 rna ligase 2 [ec:6.5.1.3]	Protein families: signaling and cellular processes	2 ± 0.6	0 ± 0	8.10
K19711	L-serine 3-dehydrogenase (nad+) [ec:1.1.1.387]	Protein families: signaling and cellular processes	2.3 ± 0.7	0 ± 0	8.10
K05824	Homoisocitrate dehydrogenase [ec:1.1.1.87]	Carbohydrate metabolism	1.5 ± 0.2	0 ± 0	8.03
K12514	Tight adherence protein f	Protein families: metabolism	1.6 ± 0.6	0 ± 0	-8.24
K13650	Mqsr-controlled colanic acid and biofilm protein a	Unclassified: genetic information processing	1.9 ± 0.8	0 ± 0	-8.26
K12147	Acidic protein msyb	Carbohydrate metabolism	1.7 ± 0.7	0 ± 0	-8.27
K02465	General secretion pathway protein s	Protein families: genetic information processing	1.7 ± 0.3	0 ± 0	-8.31
K07470	DNA gyrase inhibitor	Environmental Information Processing	2.4 ± 0.7	0 ± 0	-8.31
K12513	Tight adherence protein e	Unclassified: metabolism	1.9 ± 1	0 ± 0	-8.36
K05839	Haemolysin expression modulating protein	Environmental Information Processing	1.3 ± 0.4	0 ± 0	-8.36
K01085	Glucose-1-phosphatase [ec:3.1.3.10]	Metabolism of terpenoids and polyketides	1.9 ± 0.6	0 ± 0	-8.45
K02681	Prepilin peptidase dependent protein c	Protein families: signaling and cellular processes	2.1 ± 0.6	0 ± 0	-8.65
K07781	Luxr family transcriptional regulator, capsular biosynthesis positive transcription factor	Energy metabolism	2.3 ± 0.3	0 ± 0	-8.82
K19775	Gntr family transcriptional regulator, hexuronate regulon transcriptional repressor	Environmental Information Processing	2.3 ± 0.9	0 ± 0	-9.12
K12512	Tight adherence protein d	Protein families: signaling and cellular processes	1 ± 0.1	0 ± 0	-9.21
K08476	Two-component system, ntrc family, phosphoglycerate transport system response regulator pgta	Protein families: signaling and cellular processes	2.2 ± 0.1	0 ± 0	-9.87

K00885	N-acylmannosamine kinase [ec:2.7.1.60]		1.8 ± 0.3	0 ± 0	-9.95
K09470	Gamma-glutamylputrescine synthase [ec:6.3.1.11]	Unclassified: genetic information processing	1.3 ± 0.8	0 ± 0	-10.16
K01731	Pectate disaccharide-lyase [ec:4.2.2.9]	Protein families: signaling and cellular processes	1.3 ± 0.6	0 ± 0	-10.50

Supplementary Table 5: Chemical characterization of real municipal wastewater following primary treatment (Fundación CENTA, Carrión de los Céspedes, Seville, Spain).

	Concentration (mg L^{-1})
COD	661.00
Total nitrogen	82.00
Chlorine	141.02
Nitrite	0.20
Nitrate	1.00
Phosphate	18.62
Sulphate	19.82
Sodium	103.95
Ammonia	84.15
Potassium	41.94
Calcium	36.92
Magnesium	18.54

Supplementary Table 6: BOD and COD concentration of municipal wastewater over time during storage at refrigerated conditions (4 ± 2 ºC). Data are represented as mean of three technical replicates ± SD.

Day	**COD**	**BOD**	**BOD/COD**
0	653 ± 27	165 ± 12	0.25 ± 0.03
8	548 ± 35	132 ± 14	0.24 ± 0.04
17	474 ± 6	90 ± 4	0.19 ± 0.01